

AquaBEV: Monocular Underwater BEV Occupancy with 3D Sonar Supervision

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Abstract

Autonomous underwater robots are widely used for exploration, monitoring, and inspection, where safe navigation depends on understanding the surrounding free and occupied space. Bird’s eye view (BEV) occupancy provides such a representation, but predicting it from a single underwater RGB image is difficult due to limited, unreliable geometric cues from appearance alone. 3D imaging sonar offers complementary geometric measurements to supervise this task.

We introduce **AquaBEV**, a monocular underwater occupancy model that predicts local BEV occupancy from a single RGB image, using paired 3D imaging sonar as geometric supervision during training. **AquaBEV** maps visual features into a calibration free polar representation and applies causal decoding along the range dimension before reconstructing the prediction in Cartesian BEV coordinates. A controlled underwater occupancy benchmark was established, adapting representative occupancy methods to the same RGB to sonar task under a unified protocol. **AquaBEV** achieves 31.4 Visible IoU and 38.6 Observed IoU, 4.0% and 4.3% relative improvements over the strongest transferred baseline.

1. Introduction

Underwater environments remain among the most challenging domains for autonomous robotic operation, with applications spanning exploration, ecological monitoring, and infrastructure inspection [20, 21, 32]. Recent deployments demonstrate increasingly capable autonomy, including autonomous mapping of coral reef biodiversity hotspots [20] and cooperative surface underwater vehicle operation [29]. However, these advances hinge on reliable navigation in unstructured, GPS denied environments with degraded visual observations [33].

Reliable navigation requires a robot to understand the spatial arrangement of surrounding structures and distinguish occupied from navigable space. Recent frameworks

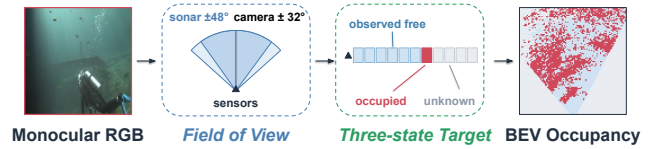


Figure 1. A single RGB image is used to predict local BEV occupancy.

such as DREAM [27] and CORAL [28] address this by maintaining a persistent BEV occupancy map as a shared representation between perception, reasoning and navigation. Unlike voxel or point based alternatives, Bird’s eye view (BEV) occupancy organizes free, occupied, and unknown space in a single ground plane frame that can be directly consumed by downstream planning [6].

Constructing an accurate BEV occupancy underwater is extremely challenging. Cameras provide rich visual and semantic information but are strongly affected by attenuation, scattering, illumination variation, and limited visibility. Acoustic sensors provide complementary geometric observations but are typically sparse and difficult to fuse with camera due to different sensing characteristics. Current systems sidestep this by relying on monocular depth estimation alone, CORAL [28] being a representative example, but the resulting occupancy maps inherit the limitations of the depth estimate, which degrades under changing visual conditions.

This motivates learning BEV occupancy directly from visual observations, without depth as an intermediate step. Using both sensors at inference increases hardware and integration complexity. Instead, sonar provides geometric supervision during training while the model predicts occupancy from RGB alone at inference. The architecture therefore captures sonar’s ordered range structure during learning, as shown in Fig. 1.

Existing BEV and occupancy methods provide several possible foundations, including learned BEV queries [18], dense monocular scene completion [3], voxel representations [17, 35], and sparse geometric primitives [12]. How-

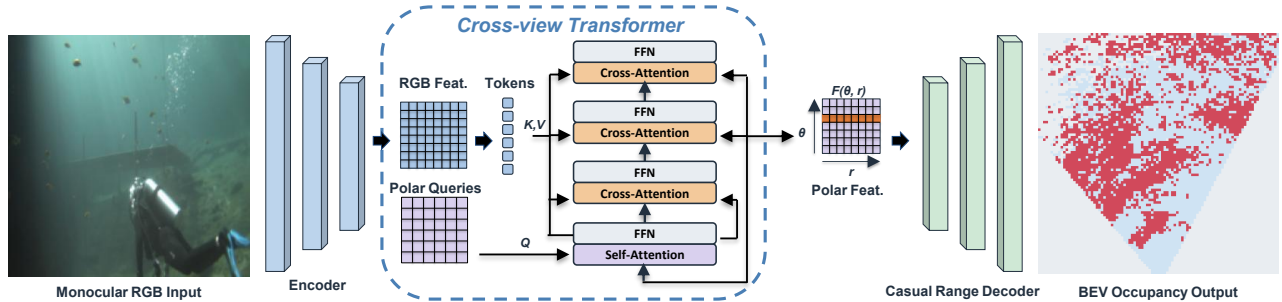


Figure 2. **Overview of AquaBEV.** AquaBEV encodes a single RGB image into a learned polar representation (θ, r) , applies causal reasoning along the range dimension, and differentially transforms the resulting polar occupancy prediction into Cartesian BEV coordinates (x, y) .

ever, these methods were designed for terrestrial settings and do not explicitly model range ordered sonar supervision. Their transfer to underwater imagery, sparse acoustic targets, and uncertain sensor geometry therefore remains unclear. We investigate this gap through a controlled benchmark under a unified evaluation protocol.

Beyond benchmarking existing methods, we design AquaBEV to reflect the native organization of sonar measurements, as shown in Fig. 2. Imaging sonar records observations by azimuth and range, giving each azimuth a natural near to far order. Rather than directly decoding occupancy in Cartesian space, AquaBEV constructs a learned polar representation with causal range decoding to model the ordered range dimension. The predicted polar occupancy is then differentially transformed into Cartesian BEV coordinates for supervision and evaluation. To the best of our knowledge, this is the first work to study monocular BEV occupancy in the underwater domain. We introduce our contributions as follows:

- We formulate monocular underwater BEV occupancy using 3D imaging sonar as training supervision and establish the first controlled benchmark for this setting. We adapt representative occupancy methods under a unified training and evaluation protocol.
- We introduce **AquaBEV**, the first monocular BEV occupancy model designed for underwater perception. AquaBEV uses a calibration free polar representation with causal range decoding and outperforms the strongest transferred baseline by 4.0% in Visible IoU and 4.3% in Observed IoU.

2. Related Work

2.1. Underwater Navigation and Spatial Mapping

Autonomous underwater navigation is challenging because GPS signals do not propagate underwater. Without GPS, AUV navigation relies on inertial estimates, acoustic positioning, geophysical information, and SLAM [23], with vi-

sual odometry and visual SLAM further reducing drift [7, 33]. Perception has expanded these capabilities through adaptive exploration [8], active visual SLAM [15], uncertainty aware inspection planning [4], and autonomous mapping of biodiversity hotspots [20]. Learned caveline detection [31], semantic visual servoing [1], and integrated perception and control [9] further connect visual understanding to underwater navigation.

Recent systems also maintain persistent spatial representations for planning. DREAM combines perception and spatial reasoning for underwater monitoring [27], while CORAL uses occupancy for local planning and collision avoidance [28]. However, these representations generally rely on SLAM, geometric sensing, or estimated depth. Dense occupancy prediction directly from monocular underwater imagery remains largely unexplored.

2.2. BEV and Occupancy Prediction

Terrestrial BEV methods transform image features using explicit geometry, as in Lift Splat Shoot [24], or learned queries, as in BEVFormer [18]. Dense occupancy methods extend this idea through monocular scene completion [3], sparse voxel queries [17], and tri planar representations [10]. To reduce volumetric cost, SparseOcc uses sparse representations [25], COTR compresses volumetric features [19], and GaussianFormer models occupancy with semantic Gaussian primitives [12, 13]. SelfOcc instead learns geometric structure from video based rendering supervision [11].

These methods balance geometric structure, representational density, and efficiency [10, 12, 25]. However, they were developed for terrestrial imagery, established camera geometry, and large datasets. Their transfer to degraded underwater imagery, sparse sonar supervision, and uncertain sensor geometry remains unclear. Cartesian representations are also poorly matched to supervision produced by a radial sensor.

2.3. Polar and Range Structured Representations

Polar and cylindrical representations preserve the radial organization of bearing and range measurements. PolarNet addresses range dependent LiDAR density through polar BEV partitioning [34], Panoptic PolarNet extends this representation to panoptic segmentation [36], and Cylinder3D preserves three dimensional structure through cylindrical partitioning [37].

Because angular cells cover larger physical regions with increasing range, polar grids introduce scale and feature distortions. PolarStream applies range aware feature correction [5], PolarFormer uses multiscale polar features [14], and PARTNER realigns features according to range and heading [22]. PVP addresses polar occupancy distortion and cross modal misalignment [30], while range conditioned convolution [2] and RangeFormer [16] adapt processing to range view geometry.

These methods primarily operate directly on LiDAR measurements for segmentation, detection, or occupancy prediction [5, 16, 22, 30, 34]. AquaBEV instead uses polar space as an intermediate representation for predicting sonar supervised occupancy from a perspective RGB image.

3. Methodology

3.1. Task Formulation and Sonar Supervision

Given an underwater RGB image ($I_t \in \mathbb{R}^{H \times W \times 3}$), AquaBEV predicts a local occupancy map in the sonar coordinate frame:

$$f_\phi(I_t) = \hat{B}_t. \quad (1)$$

During training, I_t is paired with 3D imaging sonar returns

$$\mathcal{P} * t = (x_i, y_i, z_i) * i = 1^N, \quad (2)$$

which are used only to construct supervision. Sonar measurements are not provided to the network at inference.

Sonar returns within a short temporal window are expressed in the reference frame at time (t) and projected onto the horizontal plane:

$$(x_i, y_i, z_i) \mapsto (x_i, y_i). \quad (3)$$

Each projected return marks an occupied cell. Under the first return assumption, cells traversed by the corresponding ray from the sonar origin are marked as observed free. All remaining cells are unknown because the sonar provides no evidence that they are free or occupied.

We separate binary occupancy from observation validity. Let $(Y_t(x, y) \in 0, 1)$ denote free or occupied space and define

$$M_t(x, y) = \begin{cases} 1, & (x, y) \text{ is acoustically observed,} \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

Unknown cells therefore have $(M_t(x, y) = 0)$ and are excluded from training. Figure 1 summarizes this training and inference formulation.

3.2. AquaBEV Overview

As shown in Fig. 2, AquaBEV predicts Cartesian occupancy through an intermediate polar representation. A visual encoder first extracts image features. Learned polar queries then aggregate these features into a latent field indexed by azimuth and range. A causal decoder processes the field from near to far, after which a differentiable transformation maps the polar prediction into Cartesian BEV:

$$I_t \rightarrow F_{\text{img}} \rightarrow F_{\text{polar}} \rightarrow H_{\text{polar}} \rightarrow \hat{B}_t. \quad (5)$$

AquaBEV does not consume explicit camera to sonar extrinsic parameters. Instead, it learns the mapping from image features to the sonar aligned occupancy frame from paired training supervision. The formulation therefore avoids explicit extrinsic calibration, although the learned mapping remains tied to the sensor configuration represented during training.

3.3. Cross View Transformer

AquaBEV uses a cross view transformer to map perspective image features into a polar latent representation. A ConvNeXt B encoder first extracts a visual feature map from I_t . The features are flattened and projected into image tokens

$$T_{\text{img}} \in \mathbb{R}^{N_{\text{img}} \times C}, \quad (6)$$

where N_{img} is the number of tokens and C is the latent dimension.

The polar representation is initialized using one learned query for each azimuth and range location. We use $N_\theta = N_r = 96$, covering azimuths from -48° to 48° and ranges from 0.4 m to 10 m. Range positional encoding is added to distinguish locations associated with different distances.

$$\text{CrossAttn} \left(Q^{(\ell-1)}, T_{\text{img}}, T_{\text{img}} \right). \quad (7)$$

Through repeated refinement, the transformer learns the correspondence between perspective image content and locations in the polar prediction space.

The refined queries are reshaped into the dense polar field

$$F_{\text{polar}} \in \mathbb{R}^{N_\theta \times N_r \times C}, \quad (8)$$

which is passed to the causal range decoder.

3.4. Causal Range Decoder

A conventional two dimensional decoder treats azimuth and range symmetrically. AquaBEV instead models range as an ordered dimension. For azimuth (θ), let

$$F_\theta = [F_{\theta,1}, \dots, F_{\theta,N_r}] \quad (9)$$

Method	Design		Occupancy		Range				Geometry	
	Representation	Geometry	Visible $\uparrow$	Observed $\uparrow$	Near $\uparrow$	Mid $\uparrow$	Far $\uparrow$	R Macro $\uparrow$	bF1@2 $\uparrow$	Chamfer $\downarrow$
SurroundOcc [26]	Dense voxel	Projection based	29.8	34.9	38.9	<u>40.6</u>	18.9	32.8	48.4	<u>0.355</u>
BEVFormer [18]	BEV queries	Projection based	28.7	34.6	39.7	39.5	18.4	32.6	45.1	0.398
VoxFormer [17]	Sparse voxel	Projection based	29.2	35.7	38.8	39.6	19.7	32.7	47.0	0.371
MonoScene [3]	Dense voxel	Projection based	29.8	35.8	40.8	39.7	20.3	33.6	45.4	0.381
TPVFormer [10]	Tri perspective	Projection based	29.0	34.2	37.4	39.3	18.9	31.8	<u>47.2</u>	0.382
OccFormer [35]	Dual path voxel	Projection based	28.8	36.0	38.5	40.0	19.6	32.7	46.4	0.378
GaussianFormer [†] [12]	Sparse Gaussian	Calibration free	<u>30.2</u>	<u>37.0</u>	<u>41.9</u>	39.6	<u>21.4</u>	<u>35.0</u>	41.3	0.412
AquaBEV (ours)	Polar occupancy	Calibration free	31.4	38.6	43.0	42.1	22.2	35.8	45.9	0.340

Table 1. **Comparison of monocular underwater BEV occupancy methods on the proposed benchmark.** All methods use the same RGB input, three state sonar supervision, data split, ConvNeXt B backbone, training schedule, and evaluation protocol (**BEST/SECOND BEST**).

denote latent features ordered from near to far. The decoder output at range (r) depends only on the current and nearer polar features:

$$H_{\theta,r} = \mathcal{D}(F_{\theta,1:r}). \quad (10)$$

Within the polar decoder, this constraint requires

$$\frac{\partial H_{\theta,r}}{\partial F_{\theta,r'}} = 0, \quad r' > r. \quad (11)$$

We implement $\mathcal{D}$ using four gated causal convolution blocks along range, with dilation rates ($d \in 1, 2, 4, 8$). Lightweight angular interaction additionally exchanges information between neighboring azimuth bins at the same range. Azimuth therefore captures relationships between nearby viewing directions, while range preserves directional ordering from the sensor origin.

The decoder produces

$$H_{\text{polar}} \in \mathbb{R}^{N_\theta \times N_r \times C}, \quad (12)$$

which is converted into polar occupancy logits by the prediction head.

3.5. Polar to Cartesian Reconstruction

Let

$$L^{\text{polar}} \in \mathbb{R}^{N_\theta \times N_r} \quad (13)$$

denote the predicted polar logits. A Cartesian BEV location (x, y) corresponds to

$$r(x, y) = \sqrt{x^2 + y^2}, \quad \theta(x, y) = \text{atan2}(y, x). \quad (14)$$

We transform the prediction into Cartesian coordinates using differentiable sampling:

$$L^{\text{BEV}}(x, y) = \mathcal{W}(L^{\text{polar}}, \theta(x, y), r(x, y)). \quad (15)$$

Gradients from the Cartesian objective propagate through ($\mathcal{W}$) into the polar decoder. AquaBEV is therefore optimized for the final BEV task without requiring independently rasterized polar labels.

3.6. Training Objective

The Cartesian logits are supervised only at acoustically observed cells. Given a per cell binary occupancy loss (ℓ_{occ}), the objective is

$$\mathcal{L}^*_{\text{occ}} = \frac{\sum_{x,y} M_t(x, y) \ell_{\text{occ}}(L^{\text{BEV}}(x, y), Y_t(x, y))}{\sum_{x,y} M_t(x, y)}. \quad (16)$$

The complete model is trained through this Cartesian occupancy objective. No polar labels or sonar measurements are provided as network inputs. At inference, AquaBEV maps a single RGB image directly to local Cartesian BEV occupancy.

4. Experiments

4.1. Dataset and Evaluation Protocol

We evaluate monocular underwater BEV occupancy using paired RGB and 3D imaging sonar collected across seven underwater sessions. The complete dataset contains 110 scenes and approximately 96,000 paired observations. We use sessions $\{2, 3, 6, 7, 8\}$ for training, session 5 for validation, and session 4 for testing. The split is fixed for all experiments so that every method is evaluated on the same unseen session.

For every RGB frame, the corresponding occupancy target is constructed from accumulated 3D imaging sonar measurements using the three state formulation described in Sec. 3.1. Each target uses 16,384 sonar points and contains occupied, observed free, and unknown cells. The same target generation procedure is used for all benchmark models and AquaBEV. At inference, every method receives only a single RGB image. Table 1 reports results under this fixed protocol.

We evaluate occupancy using complementary overlap, range, and geometric metrics. **Visible IoU** measures oc-

Method	Vis. $\uparrow$	Obs. $\uparrow$	Near $\uparrow$	Mid $\uparrow$	Far $\uparrow$	R Macro $\uparrow$	bF1@2 $\uparrow$	Chamfer $\downarrow$
Cartesian decoder	30.8	37.4	42.4	42.0	20.7	35.0	47.7	0.351
Polar decoder	31.3	37.5	42.3	42.0	21.5	35.2	48.0	0.342
Polar + Survival supervision	31.3	38.5	42.6	42.0	21.7	35.7	45.8	0.344
Polar + Metric angular coupling	31.0	38.4	42.5	42.0	21.5	35.3	46.2	0.349
Polar + Full range context	30.9	38.6	41.1	42.0	22.1	35.0	44.1	0.360
AquaBEV	31.4	38.6	43.0	42.1	22.2	35.8	45.9	0.340

Table 2. **Ablation of AquaBEV on occupancy and geometric quality.** All results are averaged over three seeds.

Encoder	Params.	Val.	Vis. $\uparrow$	Near $\uparrow$
DINOv2 S	23.2M	30.6	28.2	39.0
ResNet 50	25.6M	33.9	32.7	45.3
ConvNeXt T	29.5M	33.5	30.9	42.6
ResNet 101	44.6M	33.6	30.1	42.1
ConvNeXt S	51.2M	33.5	29.1	40.1
ResNet 152	60.3M	33.8	31.3	43.9
DINOv2 B	87.3M	29.3	28.0	38.7
ConvNeXt B	89.4M	34.9	32.9	45.3

Table 3. **RGB encoder ablation.** All encoders are evaluated using the same calibration free occupancy framework, training protocol, and held out test session. The results are from one seed.

cupied cell overlap across the camera visible portion of the BEV. Because this region can contain cells without acoustic observation, predictions in unknown cells are counted as false positives. **Observed IoU** instead restricts evaluation to occupied and observed free cells using the observation mask (M_t). This metric excludes unknown cells and therefore directly measures performance over supervised space.

To evaluate occupancy as a function of forward distance, we divide the visible BEV into three bands along the forward x axis. Near covers $x \in [0.5, 3.0)$ m, Mid covers $x \in [3.0, 6.0)$ m, and Far covers $x \in [6.0, 10.0)$ m. Each band is intersected with the camera visible field of view before evaluation. These intervals measure forward distance rather than polar slant range, which avoids conflating the evaluation bands with the radial coordinate used internally by AquaBEV.

We report IoU independently within each band and define **Range Macro** as the equally weighted mean

$$\text{RMacro} = \frac{\text{IoU}_{\text{near}} + \text{IoU}_{\text{mid}} + \text{IoU}_{\text{far}}}{3}. \quad (17)$$

The metric therefore gives equal importance to near, mid, and far occupancy despite their different spatial widths. The small interval $x \in [0.4, 0.5)$ m is excluded from the three distance bands but remains included in Visible and Observed IoU.

Occupancy overlap alone does not fully describe geometric quality. We therefore report **boundary F1** with a tolerance of two BEV cells and **Chamfer distance** between predicted and target occupancy boundaries. Higher IoU and boundary F1 indicate better performance, while lower Chamfer distance indicates better geometric agreement.

All models use ConvNeXt B with a latent dimension of (192) and are trained for (30) epochs with batch size (24). We use the AdamW optimizer with initial learning rate of (3×10^{-4}), weight decay of (1×10^{-4}), cosine scheduling, BEV flipping, and MixUp. Checkpoints are selected using validation Visible IoU, and predictions are thresholded at (0.5). Transferred methods use seed (0), while AquaBEV and its primary ablations report the mean and sample standard deviation over three seeds.

4.2. Comparison with Existing Occupancy Methods

Table 1 compares AquaBEV with seven representative occupancy methods under the unified training and evaluation protocol. AquaBEV achieves 31.4 Visible IoU and 38.6 Observed IoU, compared with 30.2 and 37.0 for GaussianFormer, the strongest transferred baseline on these metrics.

Across forward distance, AquaBEV obtains 43.0 Near IoU, 42.1 Mid IoU, and 22.2 Far IoU. It achieves the strongest overall Range Macro among the evaluated methods.

AquaBEV also achieves the lowest Chamfer distance at 0.340 m. SurroundOcc obtains the highest boundary F1 at 48.4, compared with 45.9 for AquaBEV. These results show that AquaBEV achieves the strongest overall occupancy performance while remaining competitive in geometric quality.

4.3. AquaBEV Ablation

Encoder selection. Table 3 reports an encoder study. ConvNeXt B achieves the highest validation and test Visible IoU and matches the highest Near IoU. A subsequent three seed comparison also favors ConvNeXt B over ResNet 50, with (32.6 ± 0.34) versus (31.7 ± 0.82) Visible IoU. We therefore use ConvNeXt B for all final experiments.

Representation and range decoder. Table 2 shows



Figure 3. **Qualitative comparison of monocular underwater BEV occupancy.**

that changing from Cartesian to polar decoding provides a modest improvement, increasing Visible IoU from (30.8) to (31.3). Causal range decoding produces the main reproducible gain, increasing Observed IoU from (37.5) to (38.6), with a paired three seed improvement of (1.10 ± 0.20) points. Survival supervision, metric angular coupling, and full range context provide no consistent improvement. AquaBEV also achieves the lowest Chamfer distance at (0.340) m.

4.4. Qualitative Results

Figure 3 compares the Cartesian occupancy predictions of AquaBEV and the transferred baselines with the sonar derived ground truth. AquaBEV produces more continuous occupied regions and better preserves their spatial extent across near and distant ranges. The improvements are most visible for elongated structures and separated occupancy re-

gions, where competing methods often produce fragmented predictions, miss portions of the structure, or introduce isolated false positives. These observations are consistent with AquaBEV’s higher Visible, Observed, and Far IoU.

Challenges remain when structures are visually indistinct, weakly illuminated, or only sparsely observed by sonar. In these regions, AquaBEV may underestimate occupied extent, merge nearby structures, or fail to recover distant occupancy. Such errors reflect the inherent ambiguity of inferring sonar supervised geometry from a single RGB image, particularly when appearance provides limited evidence of distance or occluded structure.

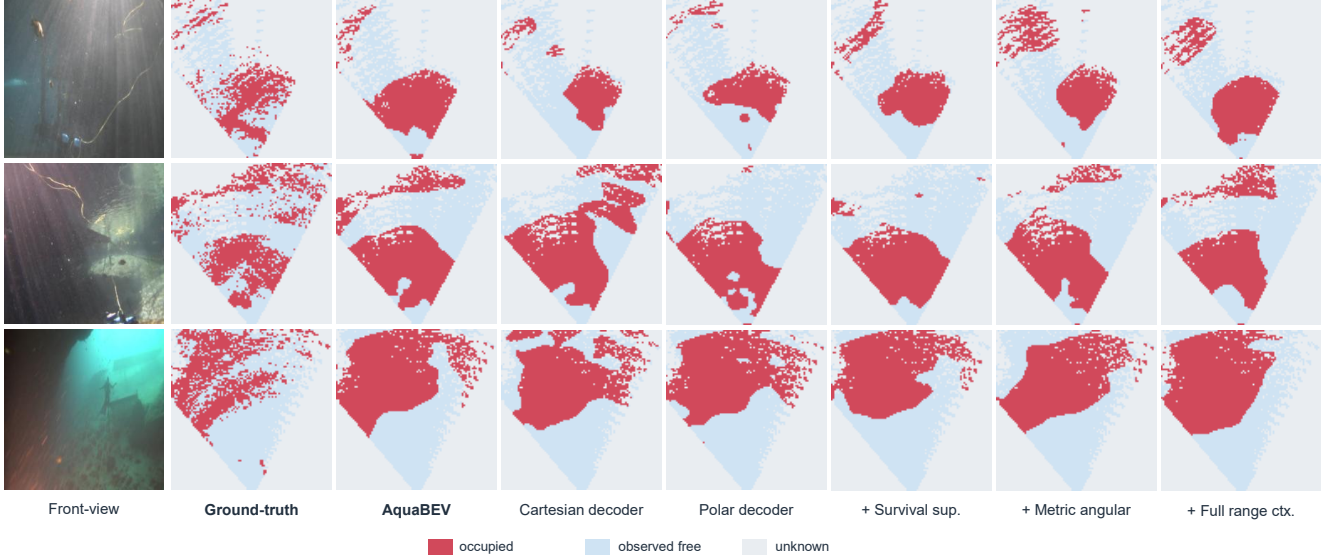


Figure 4. Ablation comparison of Casual Range Decoder.

5. Discussion

5.1. Occupancy Transfer to Underwater Perception

The benchmark reveals a notable characteristic of underwater monocular occupancy prediction. Although the transferred methods use substantially different representations, their performance remains relatively concentrated. Visible IoU spans only 28.7 to 30.2 across the seven transferred methods, while Observed IoU ranges from 34.2 to 37.0. The narrow performance range suggests that changing the occupancy representation alone has limited impact in this setting.

A possible explanation is that the primary challenge lies in recovering reliable geometry from a single underwater image. A single underwater RGB image must infer geometry supervised by a sensing modality with substantially different visibility and measurement characteristics. Increasing the complexity of the occupancy decoder cannot recover geometric information that is weak or ambiguous in the image itself. This is particularly apparent at farther distances, where every method experiences a substantial reduction in IoU.

The benchmark also suggests that geometric assumptions should be transferred carefully. Several existing approaches were designed around environments with well characterized camera geometry and stable scene structure. AquaBEV instead learns the mapping from visual features to occupancy without using camera and sonar extrinsic parameters during feature transport. Its improvement does not establish that explicit projection is generally inferior, but it indicates that learned transport is a useful alternative when

reliable cross sensor geometry is difficult to obtain.

The differences between evaluation metrics are also informative. AquaBEV achieves the strongest occupancy overlap and lowest Chamfer distance, while SurroundOcc achieves higher boundary F1. No single metric therefore captures the complete quality of an occupancy prediction. Overlap measures how much occupied structure is recovered, while boundary metrics emphasize local contour alignment and Chamfer measures the spatial displacement between structures. Reporting these jointly is particularly important for sparse underwater scenes.

5.2. Why Causal Range Reasoning Helps

The ablation study separates two ideas that can otherwise be easily conflated: using polar coordinates and reasoning along an ordered range dimension. Polar coordinates alone provide only a small improvement over the matched Cartesian decoder. The larger and reproducible change appears only after causal range decoding is introduced. This indicates that the benefit of AquaBEV does not come simply from expressing the scene in polar coordinates, as shown in Fig. 4.

The result can be understood from the different roles of the two polar axes. Azimuth describes neighboring viewing directions, while range describes progression away from the sensor origin. Treating both dimensions identically with conventional spatial convolution ignores this distinction. AquaBEV instead allows nearby azimuth regions to interact while imposing an ordering along range, so a prediction at a given distance is constructed without information from farther locations.

The additional ablations further isolate this effect. Metric angular coupling explicitly compensates for the increasing physical width represented by an angular cell at greater distance, yet it does not improve prediction. Likewise, extending the causal receptive field across the complete range sequence provides essentially the same Observed and Far IoU as the shorter AquaBEV decoder. The improvement therefore cannot be explained by greater context alone.

Explicit survival supervision also provides no consistent gain. This is important because it limits how AquaBEV should be interpreted. The model is not learning a strict first return sonar process. After 3D sonar measurements are projected into the horizontal BEV plane, several surfaces at different heights can occupy different distances along the same direction. A simple free region followed by one acoustic return is therefore not generally consistent with the supervision target.

Taken together, the results support a narrower conclusion: the useful sonar informed inductive bias is the *ordering of range*, rather than a complete physical model of sonar propagation. This distinction allows AquaBEV to exploit structure associated with radial sensing without requiring the occupancy target to satisfy an overly restrictive acoustic model.

6. Conclusion

6.1. Summary of Findings

We studied monocular underwater BEV occupancy using offline 3D imaging sonar supervision and established a controlled benchmark for transferring representative occupancy methods to this setting. Under a unified training and evaluation protocol, AquaBEV achieves the strongest overall occupancy performance among the evaluated methods while requiring only a single RGB image at inference.

AquaBEV combines a learned polar occupancy representation with causal reasoning along range. Our ablations show that polar coordinates alone provide only a modest improvement, while causal range decoding produces the main reproducible gain. Additional survival supervision, range conditioned angular coupling, and longer range context do not further improve performance. These results indicate that ordered range reasoning is a more useful inductive bias than detailed modeling of sonar propagation.

The current study remains limited to one underwater environment and evaluates AquaBEV only as a perception model. Sonar accumulation and projection also introduce geometric and temporal ambiguities into the supervision. Improved occupancy prediction therefore does not yet establish improved navigation performance.

6.2. Future Work

We plan to integrate AquaBEV with existing underwater navigation frameworks such as CORAL [28]. Closed loop experiments will evaluate whether the predicted occupancy improves path planning, collision avoidance, and navigation efficiency in practice.

We will also extend AquaBEV from binary occupancy to semantic BEV prediction by incorporating segmentation of underwater structures, terrain, vehicles, and other relevant objects. A further direction is physics informed BEV generalization across changes in water clarity, illumination, acoustic observation density, and sensor configuration. This will examine whether underwater imaging and sonar priors can improve transfer to environments and operating conditions not represented during training.

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